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Roll No

ME-503 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Mechanical Measurement and Control
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe the different types of errors in measurement with example.
b) Discuss the least square regression analysis with examples.
2. Describe the construction and working of rotameter. Explain its advantages and disadvantages.
3. a) Explain following terms related to measurement:
i) Noise and interference
ii) Replication and repetition
b) Explain zero order, first order and second order measurement system.
4. a) What are primary, secondary and tertiary measurements? Explain with suitable examples.
b) Explain the following terms:
i) Precision
ii) Repeatability
iii) Accuracy
5. a) What is strain gauge? Explain the different types of strain gauges.
b) What is thermistor? Explain its working with suitable sketch.
6. a) Explain the working of LVDT with neat sketch.
b) Discuss the construction and working of resistance temperature detector.
7. What is Mechanical System? Write down the transfer function for various mechanical systems.
OR
Discuss in brief higher order uncertainty analysis.
8. Write short note on:(Any two)
a) Signal flow graph and its application
b) Step and ramp input
c) LVDT
